



Computational Thinking

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Frequency count method

A

4	8	1	12	10
0	1	2	3	4

n = 5

i = 0
 { i = 1
 i = 2
 i = 3
 i = 4
 i = 5 ✗

```
sum(A, n)
{
```

```

s=0;           → 1
for (i=0; i<n; i++)
{             1   n+1   n   → 2n+12
    s=s+A[i]; → n
}
return s;     → 1

```

$$T(n) = 2n + 3$$

$$O(n)$$

Finding the time complexity of algorithms (1/4)

```
for (i=0; i<n; i++)  
{  
    some statement;  
}
```



$n + 1$



n

$O(n)$

Finding the time complexity of algorithms (2/4)

```
for (i=n; i>0; i--)  
{  
    some statement;  
}
```



n

$O(n)$

Finding the time complexity of algorithms ^(3/4)

```
for (i=1; i<n; i=i+2)
{
    some statement;
}
```



$$\frac{n}{2}$$

$$T(n) = \frac{n}{2}$$

$$O(n)$$

Finding the time complexity of algorithms (4/4)

<code>for (i=0; i<n; i=++)</code>		$n + 1$
<code>{</code>		
<code> for (j=0; j<n; j=++)</code>		$n \times (n+1)$
<code> {</code>		
<code> some statement;</code>		$n \times n$
<code> }</code>		
<code>}</code>		
		$O(n^2)$

Procedure

- ❑ We will discuss the problem
- ❑ You work together in pairs
- ❑ 5 -10 minutes per problem
- ❑ We discuss the solution

Problem: Genetic database

In a genetic database, there are 64 genes associated with hay allergies, and they are stored by ascending size. When biomedical scientists discover a new drug molecule which might be used to treat such allergies, they need to find exactly which of these genes the molecule could react best with. In order to do this, they compare the size of the gene with the size of the molecule.

If the gene they are looking for is the 48th smallest in size of the 64 genes, how many times more computational steps (i.e., key comparisons) would it require to look for this gene in the list via linear search than binary search?

Solution: Genetic database

Linear search: 48 key comparisons.

Binary search:

1. $L = 0, R = 63, M = 31$
2. $L = 32, R = 63, M = 47 \rightarrow$ key is found!

Answer: $48 / 2 = 24 \rightarrow$ linear search would require 24 times the steps of binary search.

Problem:

In a genetic database, there are 64 genes associated with hay allergies, and they are stored by ascending size. When biomedical scientists discover a new drug molecule which might be used to treat such allergies, they need to find exactly which of these genes the molecule could react best with. In order to do this, they compare the size of the gene with the size of the molecule.

If the gene they are looking for is the 48th smallest in size of the 64 genes, how much more computational steps (i.e., key comparisons) would it require to look for this gene in the list via linear search than binary search?

Problem: Automatic diagnostic system

Write a Python program for an automatic diagnostic system which asks the user if they have each of the following symptoms: fever, coughs, muscle aches, stomach aches, fatigue. If the user has at least three of the above symptoms, the program should output: “The likely diagnosis is the flu”, and if not, it should output “The diagnosis is uncertain”.

Solution: Automatic diagnostic system (1/2)

STEP-BY-STEP:

- Store symptoms in list.
- Store diagnosis threshold for number of symptoms.
- Create a symptom counter.
- FOR every symptom in list:
 - Ask user if they have symptom.
 - IF they have symptom:
 - Increment symptom counter.
- IF the symptom counter is higher than the diagnosis threshold:
 - Print message of diagnosis for user.
- ELSE:
 - Print message for user that the diagnosis is uncertain.

Problem:

Write a Python program for an automatic diagnostic system which asks the user if they have each of the following symptoms: fever, coughs, muscle aches, stomach aches, fatigue. If the user has at least three of the above symptoms, the program should output: “The likely diagnosis is the flu”, and if not, it should output “The diagnosis is uncertain”.

Solution: Automatic diagnostic system (2/2)

Code

```
1 symptoms = ["fever", "coughs", "muscle aches", "stomach aches", "fatigue"]
2
3 threshold = 3
4
5 symptom_counter = 0
6
7 for symptom in symptoms:
8     experience = input("Do you experience %s?" % symptom)
9
10    if experience == "yes" or experience == "Yes":
11        symptom_counter += 1
12
13 if symptom_counter >= threshold:
14     print("The likely diagnosis is the flu.")
15
16 else:
17     print("The diagnosis is uncertain.")
18
19
```

**THANK YOU FOR YOUR ATTENTION,
YOU CAN CLAP NOW**



**IF YOU HAVE ANY QUESTIONS, PLEASE
ASK MY FRIEND GOOGLE**

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